SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)
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॥वसुधैव कुटुम्बकम्॥

CSE - Academic Year 2026-27
Java Programming Lab | Batch 2025-29

ASSIGNMENT : 17

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Batch	CSE A - 2
Class	Java Programming Lab
Academic Year & Semester	2025-29 : Semester 3
Date of Submission	

TITLE : Write a Java program to demonstrate the use of Swing and Applets.

Learning Objectives : To develop graphical user interfaces using Java Swing components.

Theory : Java Swing is a GUI toolkit used to develop graphical applications in Java. It provides components such as JFrame, JLabel, JButton, and JTextField for creating interactive user interfaces. Swing components are platform-independent and provide a rich set of features for desktop application development.

Applets were small Java programs designed to run inside web browsers. Although applets are now obsolete, they are historically important in understanding Java GUI development. Swing remains widely used for developing desktop-based graphical applications.

Exercise :

1. Create a simple Student Registration Form GUI using Swing.

```
import javax.swing.*;
import java.awt.event.*;

public class java17a {
    public static void main(String[] args) {

        JFrame frame = new JFrame("Student Registration Form");

        JLabel nameLabel = new JLabel("Name:");
        JLabel rollLabel = new JLabel("Roll No:");
        JLabel courseLabel = new JLabel("Course:");

        JTextField nameField = new JTextField();
        JTextField rollField = new JTextField();
        JTextField courseField = new JTextField();

        JButton registerButton = new JButton("Register");

        nameLabel.setBounds(50, 30, 100, 30);
        nameField.setBounds(160, 30, 180, 30);

        rollLabel.setBounds(50, 80, 100, 30);
        rollField.setBounds(160, 80, 180, 30);

        courseLabel.setBounds(50, 130, 100, 30);
        courseField.setBounds(160, 130, 180, 30);

        registerButton.setBounds(130, 190, 120, 35);

        registerButton.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {

                String name = nameField.getText();
                String rollNo = rollField.getText();
                String course = courseField.getText();
```

```
                String message =
                        "Student Registered Successfully!\n\n" +
                        "Name: " + name +
                        "\nRoll No: " + rollNo +
                        "\nCourse: " + course;

                JOptionPane.showMessageDialog(frame, message);
            }
        });

        frame.add(nameLabel);
        frame.add(nameField);

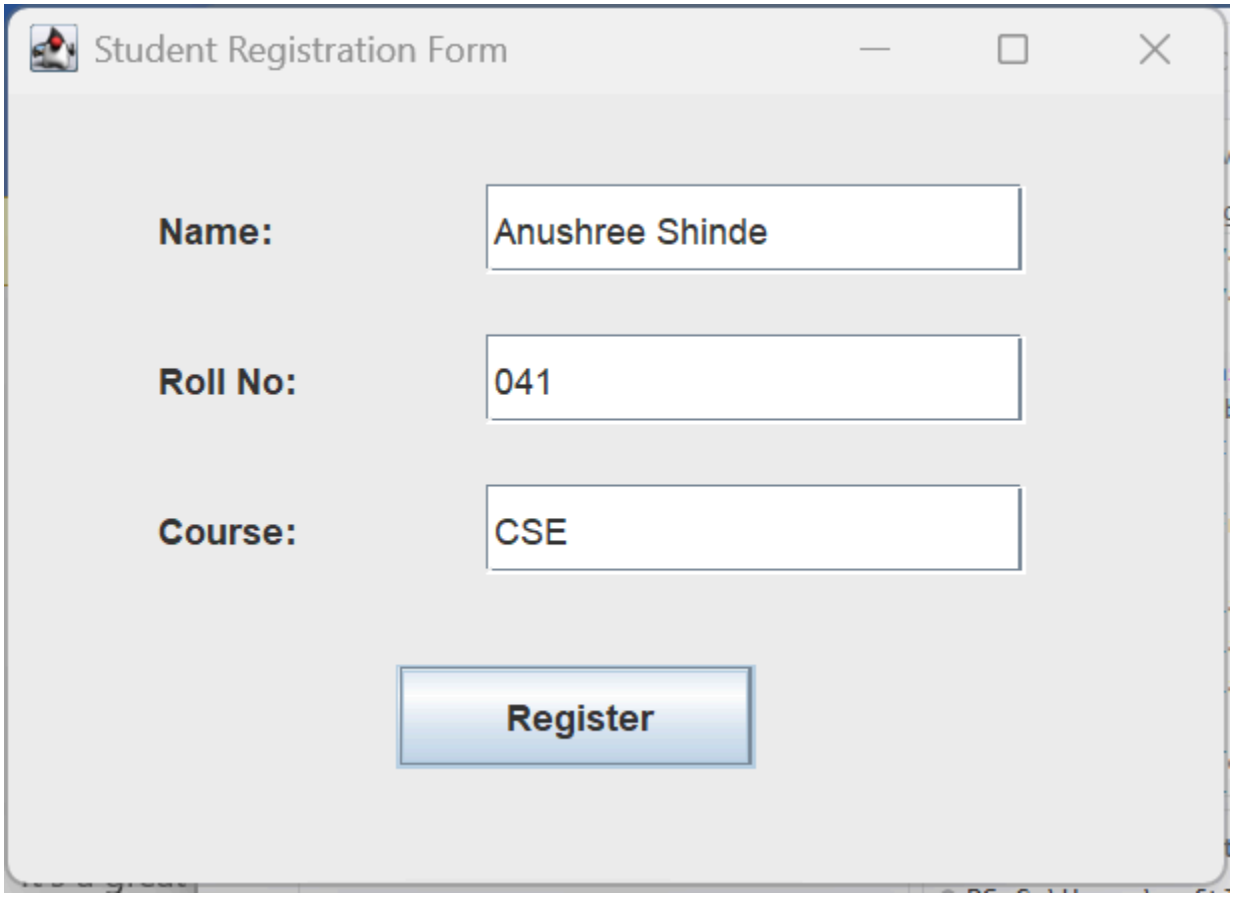
        frame.add(rollLabel);
        frame.add(rollField);

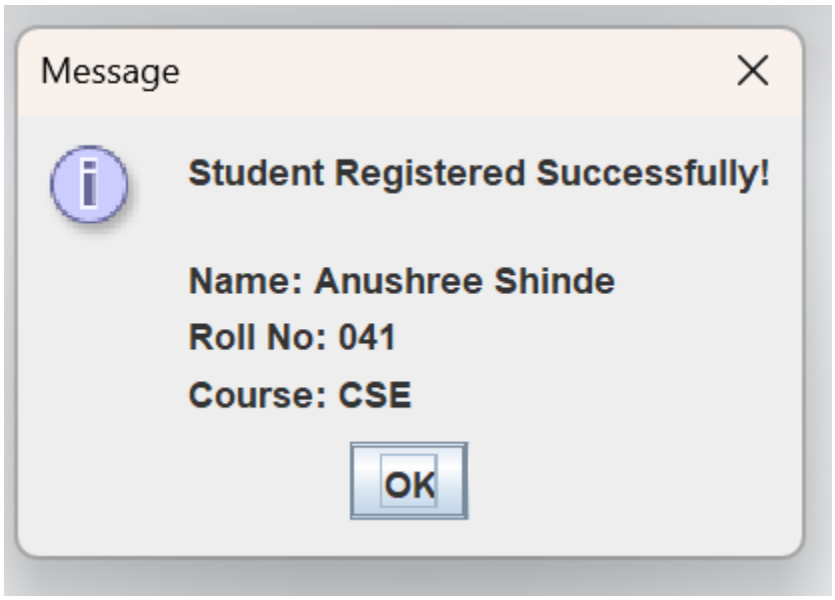
        frame.add(courseLabel);
        frame.add(courseField);

        frame.add(registerButton);

        frame.setSize(420, 300);
        frame.setLayout(null);
        frame.setVisible(true);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}
```

OUTPUT :


Student Registration Form
Name:
Anushree Shinde
Roll No:
041
Course:
CSE
Register


Message
Student Registered Successfully!
Name: Anushree Shinde
Roll No: 041
Course: CSE
OK

2. Create a simple Employee Registration Form using Java Swing GUI that takes employee details such as Employee ID, Name, Department, and Salary. Display the entered information in a dialog box.

```
import javax.swing.*;
import java.awt.event.*;

public class java17b {
    public static void main(String[] args) {

        JFrame frame = new JFrame("Employee Registration Form");

        JLabel idLabel = new JLabel("Employee ID:");
        JLabel nameLabel = new JLabel("Name:");
        JLabel deptLabel = new JLabel("Department:");
        JLabel salaryLabel = new JLabel("Salary:");

        JTextField idField = new JTextField();
        JTextField nameField = new JTextField();
        JTextField deptField = new JTextField();
        JTextField salaryField = new JTextField();

        JButton submitButton = new JButton("Submit");
        JButton clearButton = new JButton("Clear");

        idLabel.setBounds(40, 30, 120, 30);
        idField.setBounds(170, 30, 170, 30);

        nameLabel.setBounds(40, 75, 120, 30);
        nameField.setBounds(170, 75, 170, 30);

        deptLabel.setBounds(40, 120, 120, 30);
        deptField.setBounds(170, 120, 170, 30);

        salaryLabel.setBounds(40, 165, 120, 30);
        salaryField.setBounds(170, 165, 170, 30);

        submitButton.setBounds(90, 220, 100, 35);
        clearButton.setBounds(210, 220, 100, 35);
```

```
submitButton.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {

        String id = idField.getText();
        String name = nameField.getText();
        String department = deptField.getText();
        String salary = salaryField.getText();

        String message =
                "Employee Details\n\n" +
                "Employee ID: " + id +
                "\nName: " + name +
                "\nDepartment: " + department +
                "\nSalary: ₹" + salary;

        JOptionPane.showMessageDialog(
                frame,
                message,
                "Employee Information",
                JOptionPane.INFORMATION_MESSAGE
        );
    }
});

clearButton.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {

        idField.setText("");
        nameField.setText("");
        deptField.setText("");
        salaryField.setText("");
    }
});

frame.add(idLabel);
frame.add(idField);

frame.add(nameLabel);
frame.add(nameField);
```

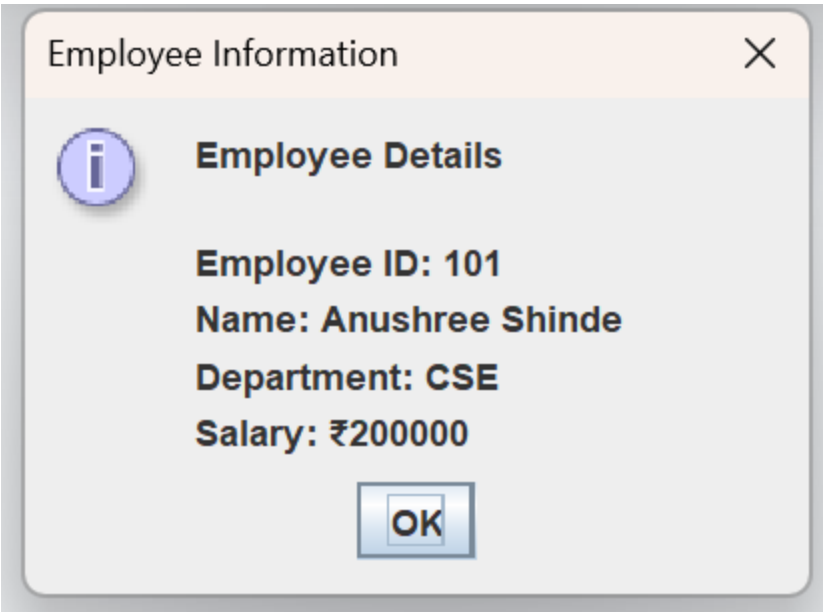
```
        frame.add(deptLabel);
        frame.add(deptField);

        frame.add(salaryLabel);
        frame.add(salaryField);
        frame.add(submitButton);
        frame.add(clearButton);

        frame.setSize(420, 330);
        frame.setLayout(null);
        frame.setVisible(true);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}
```

OUTPUT :





Outcome :

- Understood the basic concepts of Java Swing for GUI development.
- Implemented graphical forms using Swing components.
- Learned to accept and display user information through a graphical interface.

Conclusion :

This assignment demonstrated the use of Java Swing for developing graphical user interfaces. Swing components such as labels, text fields, and buttons help create interactive and user-friendly desktop applications.